

Giving BT a competitive edge through technical consultancy

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The Technical Facilities unit provides a consultancy service to major BT programmes and projects, ensuring that risk management, service integrity and whole life costing are incorporated into business plans and proposals. This paper describes some recent examples of how services provided by this unit have been used to deliver major projects and to help win business contracts.

1. Introduction

The telecommunications market is evolving rapidly. It is global and fiercely competitive, with customer and regulatory requirements demanding an almost instant response from service providers. In meeting its customers' requirements, BT must reduce both the time to market and the whole life cost (WLC) of its products and services while managing risk both to itself and to its customers.

The Technical Facilities (TF) unit at BT Laboratories (BTL) offers a portfolio of services aimed at identifying and optimising the risk taken by the business. TF consists of systems specialists with skills in risk analysis and risk-optimisation-related specialisms, and provides guidance and consultancy to a wide internal BT customer base and to BT's major partners.

This paper provides three recent examples of how TF has supported BT in its challenge of gaining new business and also in understanding risk.

- The first example explains how the use of integrated logistics support (ILS) helped BT's bid to work with the Ministry of Defence (MOD) on an advanced fixed telecommunications network for the UK armed forces (the Defence Fixed Telecommunications System (DFTS)). This network will deliver new voice, data, LAN-interconnection and other wide area working services for the Army, Navy and RAF. Of particular note is the emphasis on WLC and the need to provide complete solutions to customers.
- The second example provides an overview of the new end-to-end security process that facilitates the identification and management of security requirements of new BT products and systems. It explains the

business benefits, in terms of risk management and WLC, of this process and provides an overview of its component parts.

- The final example explains how new models have been developed to demonstrate large system compliance with the European community directive on electromagnetic compatibility (EMC). A failure to comply with this directive would have a serious impact on BT's operations. This is another example of a world class technique which minimises risk to BT systems, and hence benefits customer interests.

2. The application of ILS to the Defence Fixed Telecommunications System

2.1 Background

A consortium led by BT has bid for a contract to replace the existing MOD telecommunications networks in the United Kingdom. The collective name for these networks is the Defence Fixed Telecommunications System (DFTS).

Much of the infrastructure had been built using public telecommunications operators (PTO) bearers and support systems, but the design and systems integration had been undertaken by the individual services (i.e. Army, Navy, RAF and central staff). A project definition study had identified opportunities for significant savings in operations and support costs through the integration of the 51 existing networks. Tenders were therefore invited for a contract to create new networks and services with seamless connectivity for services personnel. A primary objective was to enable services personnel to concentrate their efforts on core activities and hence the contract favoured an 'outsourcing' approach.

Over the last decade the MOD has consistently found that equipment support costs have been higher than anticipated and that there have been difficulties with maintaining equipment. This has been true not just for weaponry (i.e. tanks, military aircraft, and ships, etc), but also for infrastructure equipment such as telecommunications. Figure 1 illustrates typical support problems, which result in additional cost in service industries. The high rate of no fault found (NFF), shortage of adequate test equipment or difficulties with packaging, handling, storage and transportation (PHS&T) are some of the common problems.

To address these problems, the MOD asked that tenders contained proposals for ILS in which minimising whole life cost and maximising operational effectiveness were the prime objectives. ILS is now a primary selection parameter in the MOD tender assessment process. It is making increasing use of ILS in new equipment programmes where reliability and maintainability are the main objectives. In order to satisfy customer requirements for ILS, which is a new concept for the telecommunications industry, TF have taken a lead in this field by providing a professional consultancy service.

Sections 2.3 and 2.4 outline the principles of ILS and how it requires tailoring for a telecommunications service. They also describe the work undertaken for DFTS to produce the ILS proposal within the BT bid.

2.2 The DFTS bid

Due to its size and complexity, the DFTS tender submission, evaluation, and selection of the preferred contractor took approximately two years. The MOD wanted a supplier who could offer value for money and who has the necessary size, infrastructure, and professional ability to deliver the new services. Because of the difficulty and expense of re-tendering it was decided to award a ten-year contract.

As a result it was necessary to fully cost the services offered, including the support costs, together with a mechanism for a price indexation scheme to ensure ongoing value for money. The new services required were:

- a circuit switched service for some 180 000 users with a private numbering scheme,

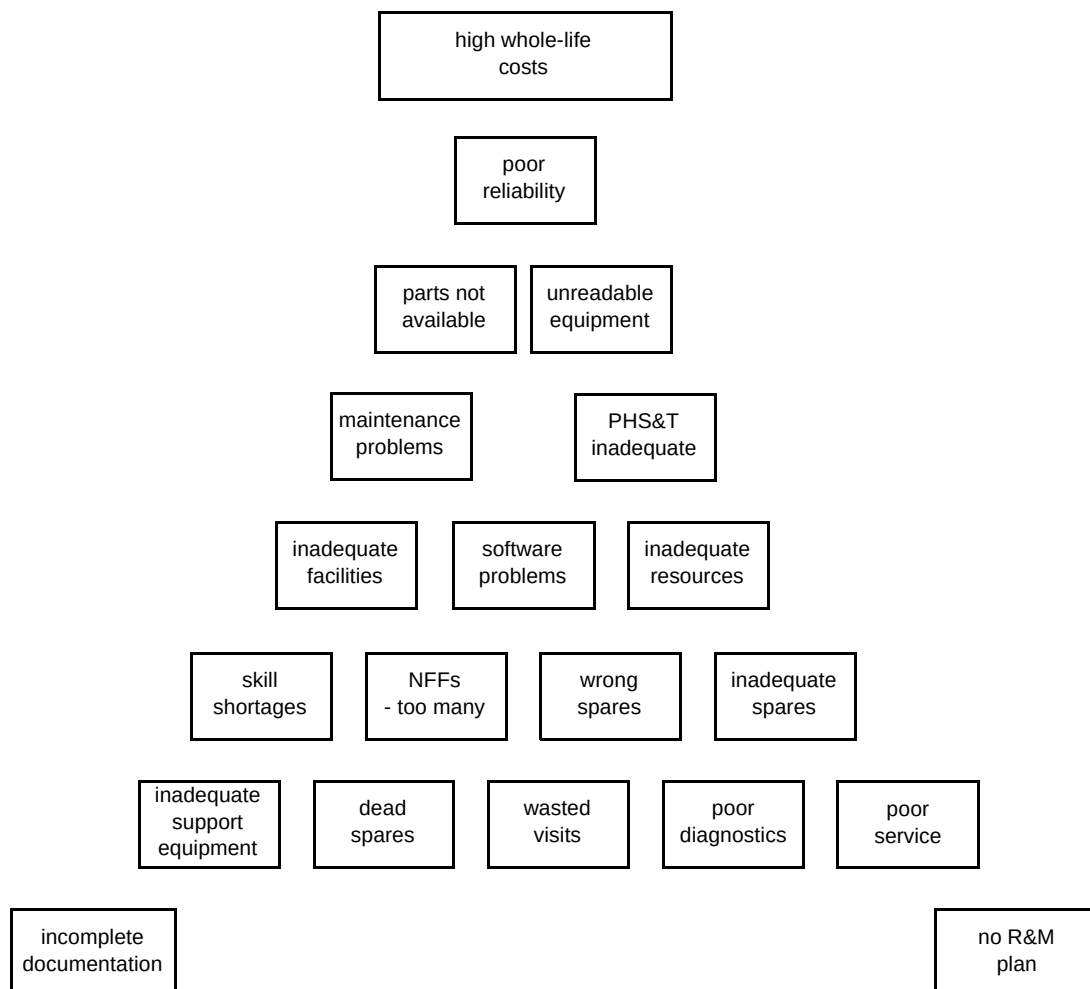


Fig 1 Mountain of support problems.

- a point-to-point service with in excess of 7000 circuits, primarily for data but also with a requirement for some voice circuits,
- a packet switched service with 50 packet switches and 2000 user ports,
- secure services.

It is not intended to discuss the detail of the BT consortium's technical proposal here, but each service required elements of standard PTO services together with new equipment and facilities to integrate the proposed designs. The costing for the proposed services was therefore based on a combination of existing tariffed services (including the maintenance of private automatic branch exchanges) and whole life cost for new equipment.

The operation and support of these services was high among the MOD's concerns, and therefore the BT consortium developed a proposal to give customer-specific operations and support services. This required a high degree of integration of the customer facing system (ServiceView) with BT's operational support systems.

2.3 The introduction and background of ILS

The term 'integrated logistics support' was introduced as a concept in the early 1970s by the United States Department of Defense to address the significant costs associated with supporting their equipment, which has been estimated at between two and three times the initial acquisition costs [1, 2]. It was important that the intended reduction in whole life cost was not to be achieved at the cost of operational effectiveness; rather, it was to be achieved in parallel with an improvement in the ability to deploy equipment successfully.

What is ILS?

ILS is a structured approach to ensure that logistic support is considered during the system design, build, and implementation on equal weighting with performance and cost. It is a disciplined, unified and interactive tool for the management and technical activities necessary to:

- integrate support considerations into systems and equipment design,
- develop support requirements that are related consistently to readiness objectives, to design, and to each other,
- acquire the required support,
- provide the required support during the operational phase at a minimum cost.

The main elements of ILS [3] are encapsulated in Fig 2.

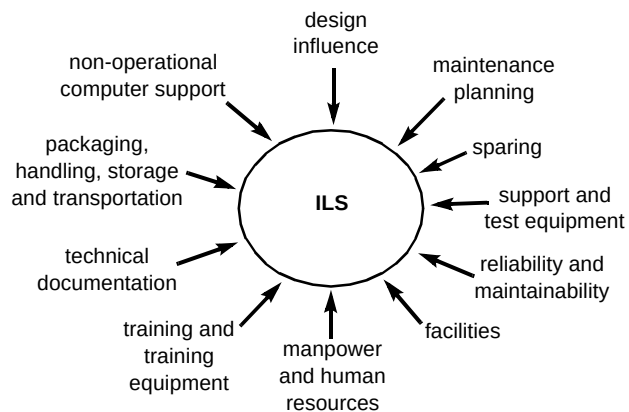


Fig 2 The elements of ILS.

In simple terms, ILS integrates the maintenance and logistics support analysis with design and applies an engineering analytical judgement, rather than rule of thumb, to the design of the final support system. Figure 3 illustrates the change from the traditional approach to logistics support.

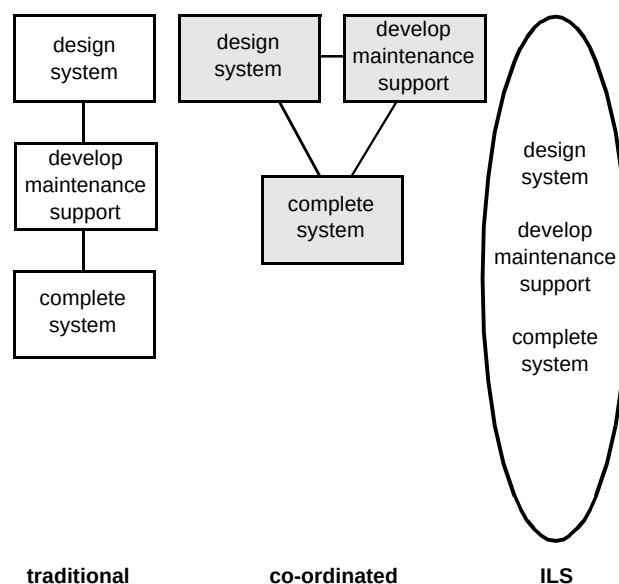


Fig 3 Approaches to design and development.

Whole life costing

The principal aim for the introduction of ILS is the reduction of the total cost of a project (termed the whole life cost) through the reduction of the support element of the whole life cost equation:

Whole life cost = acquisition cost + support cost + disposal cost

This concept is not new to telecommunications operators. For example, BT's main revenue stream is from

tariffed services such as the public switched telephone network (PSTN) and bearers such as KiloStream and MegaStream. The whole life cost of the service is calculated and then offered to customers as a combination of an initial fee, a standing charge, and a usage-based charge calculated to offset the costs and introduce a profit margin. Decisions made at early stages (such as design) based on whole life cost have long-term implications.

2.4 *The application of ILS to DFTS*

The requirement for ILS in the tender

The MOD placed a strong emphasis in their tender documentation for tenderers to integrate the support functions as well as understanding the cost base [4]. As a result the following were the main elements of the BT consortium's ILS response:

- building maintenance organisations and processes,
- specifying some of the operational support systems,
- calculating quantities of spares, planning their location and control systems,
- developing a training plan,
- identifying dependencies on the MOD,
- planning manpower for support organisations,
- identifying test equipment.

Since a number of these planning and performance modelling activities (specifically modelling of availability) were based upon an understanding of fault rates, the TF team, leading the writing of the ILS response, also undertook responsibility for responding to the performance requirements.

The bid response

For telecommunications systems and services, the following differences from a traditional ILS approach, applicable to tanks, military aircraft, and naval ships, were identified from an early stage in the development of the bid proposal:

- the largest elements of the design solutions were already in operation, hence the support infrastructure was in place,
- there was little design required at a component level,
- a number of the support processes for internal BT systems were operational,

- the basis of the proposal was a managed service in which the BT consortium would operate and support the services offered (i.e. the whole life cost was built into the tariff for each of the services).

The ILS requirement therefore needed tailoring to be directly applicable to telecommunications services. The BT consortium's ILS approach was to:

- address the support planning for the new system components,
- integrate the systems and processes for centralised reporting of faults and the presentation of service performance data (a holistic approach).

Each of these is described more fully below (note that the managed bandwidth service has been used as the vehicle for describing the logistics analysis process although it was not the only service to which it was applied).

Planning for the managed bandwidth service

The proposal for the managed bandwidth service (proposed to provide the majority of point-to-point circuits) is a new network using multiplexers. It is very similar to the existing flexible bandwidth service but is characterised by the need for additional security. This is provided by a 'stand-alone' network and the use of bulk encryption on some of the bearers. Scenarios for the geographical location of the repair service centres were evaluated and nine centres were chosen to meet the requirement for a four-hour response time for 24 hours per day and 365 days per year.

Due to its similarity with the flexible bandwidth service, network data was available for the anticipated number of faults on the network. This data facilitated the identification of manpower and spares requirements. To minimise costs associated with the multiplexer equipment, in line with the need to make a cost-effective proposal, the logistics of the spare parts deployment were to:

- locate the majority of the spares holdings at the repair service centres,
- have a centralised stock holding of spares which met the criteria of being (a) expensive and (b) rarely failing (these spares would be delivered to the required location by courier using a 24 hours delivery service),
- calculate the spares requirement using a 'stock-out risk' algorithm developed by TF.

Each of these contributed to cost savings in the procurement of the spares but the most significant was the use of the 'stock-out risk' algorithm. The calculation of the number of spares required at each location was made by a statistical analysis of the probability of spares not being

available assuming a random failure distribution. For non-random failure events (e.g. a lightning strike at a multiplexer node) the courier service would be used to deliver spares to the failure site.

The training policy was to multi-skill the maintenance personnel to provide a flexible workforce. The intention was to provide the geographical coverage and enable the engineer to travel to the failed site within an agreed three hours logistics delay time.

Centralised fault reporting and presentation of service performance data

The variety of the services offered, together with the number of agencies providing support services, led to a complex design for the fault reporting process. Nevertheless the intention was to make this seamless to the customer by providing a single point of contact for the reporting of faults. The high-level fault reporting process is shown in Fig 4.

The proposal required the migration of the existing processes and systems for fault reporting to a single process supported by integrated support tools. This would simplify the process of fault reporting, reduce costs and improve repair response. One of the objectives was immediate

restoration of DFTS service wherever possible with follow-up repairs of faulty elements as quickly as practical.

The MOD already has the benefit of a reporting centre for some of their systems. To implement the single fault reporting process, a single service centre was proposed. This required systems and processes that were integrated with those of the network control centre (also provided specifically for DFTS).

The existing process implementations for fault management were used as the baseline for the DFTS. The initial task was to capture and record the existing process implementations. A target process was then designed which would reduce both cost and risk.

The system for delivering the fault-reporting process was based upon ServiceView and specifically MCSS (major customer service system) [5]. Due to the need for security of the MOD's data the service centre operators required two MCSS terminals — one with MOD specific data and the other to enter faults on to BT core systems. These had to be physically separate terminals to meet the security requirements.

Once these basic principles had been established, the detailed process design was undertaken. The processes were based upon 'case team' working wherever practical,

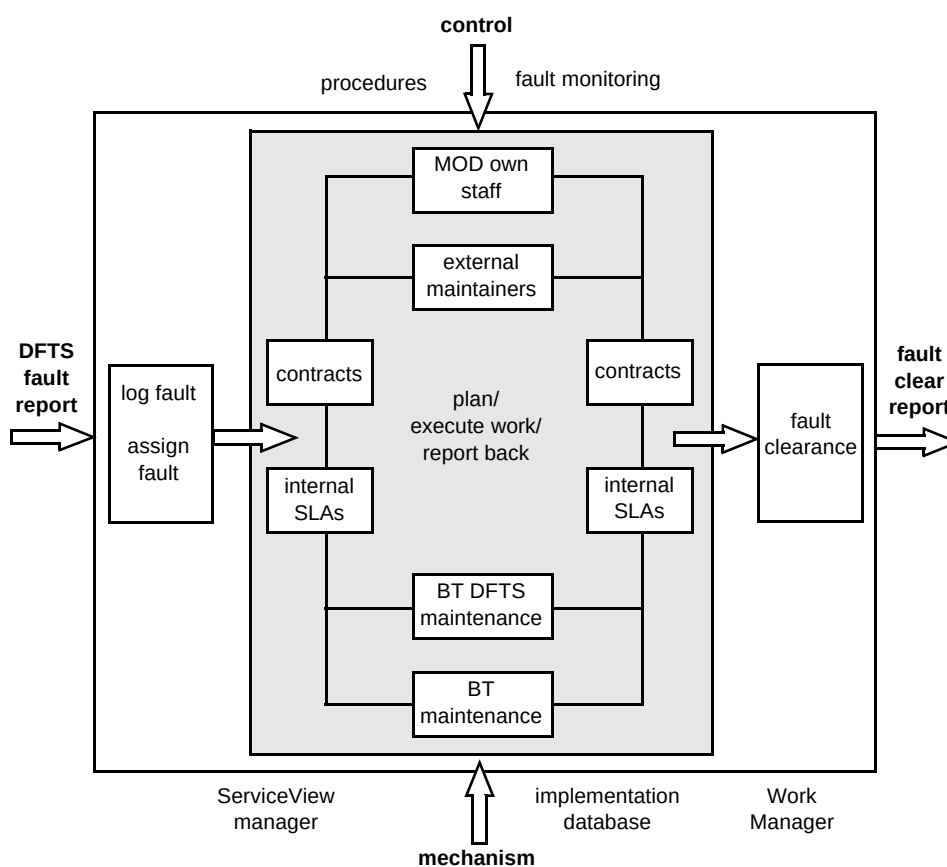


Fig 4 The centralised fault reporting process.

whereby the fault receipt, diagnosis and service restoration or placement into the appropriate fault queues (or hand-over to an external repair agent) could be performed by the customer service centre. For some services the complexity and specific knowledge required for the fault diagnosis made 'case team' working impractical.

Following the decision to make the BT consortium the 'preferred supplier' for DFTS, one of the main areas for negotiation before a contract was awarded was the service credit scheme. There were a number of elements to this, many of which related to the performance of the prime contractor management team. For the individual services the prime 'end-to-end' service measures which attracted service credits were:

- availability,
- grade of service (for switched services),
- provision times,
- fault response.

Two of these (availability and fault response) were central to the ILS proposal for the bid. The ILS team was central to the availability modelling, the logistics planning for fault response, the estimation of the risk of not meeting the performance targets, and the definition of the measures for the contract.

Environmental impact assessment of DFTS

The environmental technology unit, part of TF, conducted an environmental impact assessment (EIA) of the proposed architecture for the DFTS, based on an abridged life cycle assessment that identified the actual impacts of the proposed architecture and made recommendations to mitigate the impacts where appropriate. The report also developed BT's environmental policy that would meet BT's legal obligations during the implementation of the new architecture for DFTS.

2.5 Summary of the benefits of ILS

One of the most valuable lessons learned from the DFTS bid is the potential benefit in cost reduction of implementing ILS within the telecommunications industry.

By analysing the requirements for maintenance and logistic support early at the design stage in a structured manner, it is possible to identify major cost drivers to support the expected operational life of the equipment. ILS is a powerful tool to achieve this objective. In simple terms ILS improves system/network availability and supportability at a reduced whole life cost. ILS is a tried and tested approach, adopted by the MOD and the US Department of Defense. As the international telecommunications industry

grows more competitive, an opportunity is opening up for BT and its suppliers to implement ILS methodology on any equipment which requires maintenance support. ILS can prove to be of ongoing value within BT to assist in the building of life cycle costs for new service platforms and new products and services.

3. End-to-end security process

3.1 Overview of process

BT's ability to protect its customers and business processes from the impact of security failure will increasingly be an important market differentiator. BT must not only consider and manage threats to its customer's and its own interests, but be seen to do so using accepted and auditable processes. Simply, BT needs to understand the security requirements of each of its products and systems, implement them and be able to prove that it has done so.

Just having an accepted process in place is not enough; it must be effective. Customers will be aware of the reputation of telecommunications service providers, especially in the area of new telecommunications services, Internet and electronic commerce systems. For BT to attract and keep customers, a proven track record for protecting their interests will be a significant asset. To maintain this reputation, BT must continue to reduce its exposure to the threat of security failure and be in a strong position to detect problems and respond to them. For an individual product or system this can be achieved by defining and implementing a security policy commensurate with the likelihood and potential impact of failure.

To facilitate a company-wide implementation of security policy, BT's IT security unit (ITSU) has defined the end-to-end security process and has helped to build many major processes within it. The end-to-end process identifies the main development life cycle activities required to define, implement and maintain security policies — security policy documents (SPDs) for systems, and product protection policies (PPPs) for products. It also describes the capture and evaluation activities of the BT security evaluation and certification scheme (BTSECS) which monitors the quality and implementation of policies. Escalation procedures are built into this latter scheme to ensure that potential security problems are brought to the attention of senior managers at an early stage. Products or systems that pose a serious risk to BT will not be permitted to pass into their operational phase.

The end-to-end security process will significantly lower the risk associated with the launch of new products and systems and, by proactively addressing security during the life cycle, will help to reduce the whole life cost associated with security.

3.2 Process background

Security requirements have been well defined within BT for many years. They were expressed as corporate policy rules within the now superseded computer security manual. Compared with other organisations, the security rules were strict and comparatively well implemented. However, as the use of IT has grown, and as the technology has become more diverse and complex, the simple set of rules has become less effective; greater thought and application is now required to achieve an adequate and cost-effective level of protection.

To cater for this increased complexity, the requirement for security policy documents (SPDs) was introduced. The new system allowed policy to be defined on a system-by-system basis, allowing the system-specific vulnerabilities to be addressed by a customised set of rules. However, early methods for producing these documents were labour intensive; the SPDs themselves were large and unfocused documents and in practice were difficult to implement.

On the product side, the product protection policy (PPP) process was developed by the Networks and Systems (N&S) policy team. PPPs contain the security requirements of new BT products. This forms part of the product development and introduction process. The N&S feasibility security teams (FST) work with product teams to identify the main product security issues and ensure a smooth hand over of support to the development support units. Because products may be implemented as one or more IT systems, PPP requirements flow through to the SPD(s) of the system(s) used to deliver the product.

In 1996, working with BT Security, the ITSU developed and introduced a revised SPD development process, the output of which is a slim SPD that highlights the specifics of the target system. The SPD positions the need for security with a brief outline of function, architecture and security impact analysis summary. It states the main specific threats working against the system and the additional rules required countering them. It also contains a commitment statement to, and details of any authorised exceptions from, corporate policy. This new process has significantly reduced the effort involved in the production of SPDs and the supporting on-line documentation enabled owners to manage their production with minimal support from the BT security community.

Also in 1996, as part of a drive to improve the level of security throughout BT, the 'seamless capture' process, a simple QMS directive which works towards the registration of all products and systems, and the BTSECS, have both become mandatory.

While all the pieces of the jigsaw have been available to specify security requirements for products and systems,

there was a need to pull them together to ensure that they could be delivered effectively and at minimum cost. This would bring to developers the benefit of knowing which activities are required at each life cycle stage, how to perform them, and where to turn for support. This section explains how this is achieved, the end-to-end security process, and highlights main activities undertaken by TF security units.

3.3 Process overview

The end-to-end security process has five main logical building blocks as shown in Fig 5 and these are outlined in the following sub-sections. The detailed activities required to conform to the process have been published by the ITSU as the development security guide, within the N&S Security WWW pages accessible to internal users on the BT intranet.

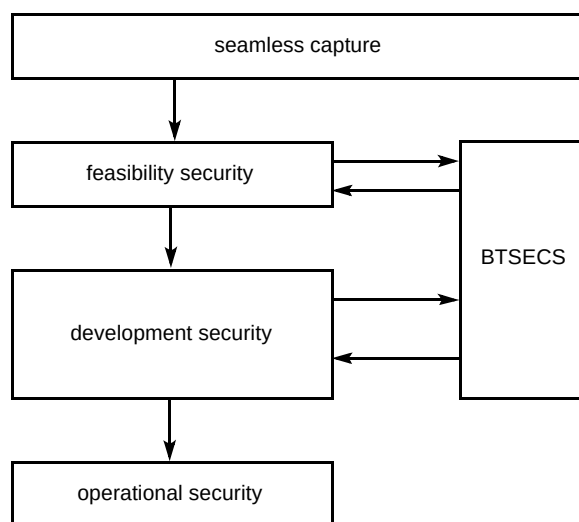


Fig 5 The block process.

Seamless capture

This is the name given to a set of QMS requirements that place responsibility on business units to register all new product and system developments with the BTSECS scheme. They also ensure that the gatekeeper roles within N&S Design and Build, and Computing Service and Operations (CSO) confirm registrations and reconcile details recorded by BT Security on the security database.

Without this information it is not possible for BTSECS to become involved at an early enough stage in the development life cycle and thus it is unable to effectively monitor and evaluate product and systems development.

Feasibility security

This stage considers security from product feasibility through to the definition of the systems required supporting

the product. This covers the capture of security requirements as stated in the BT product team handbook.

Product development starts with a feasibility study. A rapid impact analysis is conducted and a security issues document produced which identifies any high-level security requirements and concerns. These may influence high-level design options, may have major cost implications, or may be serious enough to result in the cancellation of the product.

Following feasibility, or if the feasibility stage requires more than a quick high-level look at the product and requires some development effort, the detailed product security requirements are identified and documented within a PPP. These are the high-level controls required to protect the product and its information. The PPP does not deal with the technical detail required to manage computing/network systems, but is more concerned with the potential abuse of functionality or information relating to the product itself.

Where a product is implemented as a computing/network system(s), or built on to existing systems, the PPP requirements are enforced by the component systems. The product high-level design will identify each of these component systems and map on to them the PPP security requirements that they will implement. There is often a many-to-many relationship between products and systems, i.e. a product may be implemented using several systems, and each system may support many products.

The PPP requirements must then be built into the SPDs of the component systems and implemented as development activities within the system development life cycle. Some developments will involve changes to existing systems and hence their SPDs, while others will require a new system to be designed and built. Each SPD covers the PPP requirements supported by the system and the technical detail needed to manage the component system in a secure manner.

Development security

Activities in this stage are aimed at the timely definition of security requirements, their design and implementation prior to the 'go live' date. The early definition of security requirements allows them to be designed and built into the system at an appropriate point, therefore removing the need for a costly 'bolting on' of security late in the development life cycle. During this phase, progress will be monitored by the BTSECS scheme. The development security guide identifies the detailed activities that must be undertaken during development to achieve an effective and well-managed implementation of these requirements.

Security requirements are presented in the form of an SPD. As stated in section 3.2, the SPD positions the need for security with a brief outline of function, architecture and security impact analysis summary. It states the main specific threats working against the system and the additional (non-standard) policy rules required to counter them. It also contains a commitment statement to, and details of any authorised exceptions from, the SPD benchmark. The SPD benchmark, based on corporate policy, BS7799 [6] and known best practice, is a repository of controls one would expect to see implemented in all new systems, and where practicable in all existing systems (e.g. secure operating procedures, access control, configuration management, back-ups). The four phases are described below.

- Requirements

To define the SPD is a fairly straightforward process that populates a standard pro forma. By authorising the SPD the system owner accepts any associated risk. BT Security, by co-authorising, acknowledge and approve exceptions to corporate policy. The SPD must be ready for authorisation by BTSECS at the end of requirement definition stage. At the end of the development process, the SPD will be updated to reflect implementation detail and re-authorised by BT Security.

- Design

During the design phase decisions must be made about each requirement. They will be implemented as:

- developed software/hardware solution,
- technical configuration,
- documented procedural control,
- user/support guidance document,
- documented responsibility,
- action (e.g. training, one off retrospective implementation drive).

This is documented as a design decision in the SPD implementation matrix and in the case of a software or hardware implementation in more detail in the architecture and design documents.

- Build

During this phase implementation is tracked using a document called the SPD implementation matrix. Against each policy statement it will contain information covering:

- ownership, i.e. the unit or individual assigned to implement the policy,

- high-level implementation details, i.e. design decisions, what needs to be done, what has been done,
- anticipated completion date/sign-off.

The document starts life in the early stages of development as a planning document and ends up providing traceability from policy through to the implemented control. The maintenance of this document is an important project management task.

- Systems integration

During this stage systems will pass through verification, validation and testing (VV&T). This set of activities proves that the SPD has been implemented and the solutions work. The level of proof required will depend upon the system. A strategy for proving the successful implementation of major security controls will be developed and implemented. This will be supplemented with specific (and reusable) test documentation.

Not only should testable functionality be subject to VV&T but evidence of effective implementation of all types of designed solutions, as documented in the SPD implementation matrix, must be checked for their fitness for purpose. These activities will include checking for evidence that actions actually took place, that procedures and user documentation are fit for purpose and that they meet the requirements of the SPD.

Should a system slip through the seamless capture process, or fail to fully implement the policy within the development life cycle, SPD implementation can be managed using a security uplift plan, i.e. outside the normal development life cycle.

Operational security

This is the day-to-day management of products and systems. Activities are aimed at maintaining compliance with policy, correct and continued implementation of secure operating procedures, secure configuration, security monitoring and responding to security incidents and weakness.

To help achieve this in an effective and efficient way, the ITSU will shortly be taking a major role, working with the Computer Systems unit and CSO, to drive SPD implementation through to the operational arena. Many SPD benchmark controls are implemented in the day-to-day operation and management of BT's IT architecture. The aim of this exercise will be to provide SPD-compliant architectural building blocks, subject to certain predefined conditions.

Each architectural block, e.g. Unix or Oracle, will be considered in isolation from any specific application. A technical security management standard (SMS) and secure operating procedure (SOP) set will be produced for each. The SMS will be fronted by an SPD commitment matrix and will indicate which SPD benchmark statements are implemented by the building block, firstly in consideration of the application and secondly as part of the management of the building block itself. Behind this matrix, the standard will mandate the technical and procedural control needed to implement it. Additionally it will specify any additional best practice configuration to counter known security anomalies of the target component.

Areas such as privilege management, support access and the control of support tools within the building block are issues that must be tackled at this level. This approach will have two flavours:

- common systems, e.g. MVS, which will be delivered SPD compliant — operational compliance will be conditional upon maintaining this configuration within the agreed limits specified within the SMS and compliance to a set of SOPs,
- configurable components, e.g. Unix, which will be SPD compliant when configured and managed according to the SMS and SOPs.

This will facilitate the security management of architectural building blocks which span systems, reducing the overall cost and increasing the effectiveness of security within BT's IT architecture.

BTSECS

At the time of going to press, the BTSECS scheme has just been revised. The ITSU has made a significant contribution to the re-definition of the deliverables of the scheme, making it more focused on the delivery and implementation of the SPD rather than the previous set of requirements. The scheme has been designed as an independent function. It is capable of monitoring the timely implementation of security requirements and providing an escalation route to senior management.

Following registration, owners are sent details of what will be expected from them. An evaluator will be assigned to the system and the owner will be expected to supply major milestone dates such as 'end of requirements capture', and 'end of testing'.

The first date will be used to set a date for a meeting between the system owner and the nominated BTSECS evaluator. At this meeting the SPD will be authorised by the owner and evaluator. If the SPD has not been produced, then it will be produced at this meeting. This is an escalation point. Failure to set up the meeting or agree an SPD will result in escalation.

At this meeting the confidence level model will be used to determine the assurance level of the system. This will determine how the system will be evaluated, either on a self-audit basis for low-risk systems, through to full external evaluation.

During development, the evaluator will monitor the rate of implementation and will inspect certain (dependent upon assurance level) implementation details. The evaluator needs confidence that the SPD will be implemented before the system is due to go live and that the implementation is fit for purpose. If in doubt the evaluator will seek clarification and if still not satisfied will invoke the escalation procedures.

Completed shortly before go-live date, the evaluation will have one of three outcomes:

- a full certificate will be issued,
- a provisional certificate will be issued (dependent upon a suitable security improvement plan being agreed),
- the system will fail and, therefore, will not go live until major problems have been cleared (escalation will have been invoked well in advance of this situation).

3.4 Security support

Within BT's Systems Engineering (SE), there are four main units which support development activities covering the end-to-end process and a fifth which acts as an independent facility for specialised security evaluation. Two of these units, the FST and the ITSU, are part of TF.

The key to effective support starts with the enabling of developers to take responsibility for the security of their own deliverables through 'how to' advice provided by the ITSU using documented processes such as the end-to-end process, the SPD development process, and SMSs.

This is supplemented by varying degrees of consultancy taking the form of either help in the preparation of policy, deriving security solutions, or representation on programme boards. The level of help offered is dependent upon the complexity of the product or system and of the potential impact that a failure would have upon BT's business.

Figure 6 shows the relationship of the security units within SE. They work together as a 'virtual team' ensuring

that projects are supported in the most effective manner, with people, expertise and information being pulled in to help out as required.

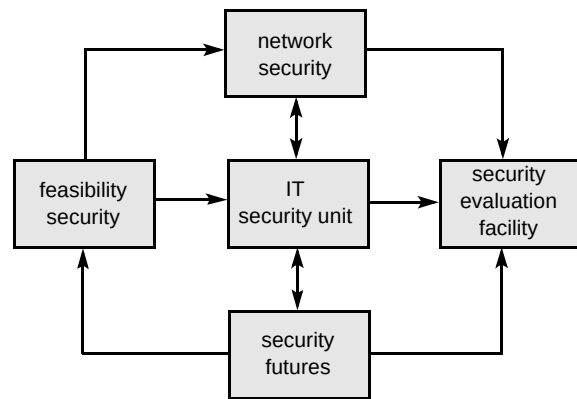


Fig 6 Systems Engineering virtual security team.

This support is supplemented with feedback from BT Security as part of the BTSECS evaluation process.

3.5 Summary of the benefits

TF has provided a major tool for the management of the business risk associated with IT systems by the development of the end-to-end security process. In doing so it has also helped to reduce the cost of an effective security implementation by ensuring that security is addressed at the correct stage in the development life cycle. The introduction of the new SPD process has reduced the complexity of SPD and significantly reduced their production costs.

The end-to-end security process will provide a framework for identifying and managing security requirements as 'business as usual' within the evolution, development and maintenance of BT products and systems.

The increasing use of the Internet for electronic commerce and the development of new telecommunications services will raise many security challenges which, if approached in a systematic way, will allow BT to forge a world-class reputation for protecting customers from the impact of fraud, loss of business and embarrassment.

The end-to-end security process supported by the TF security units will help BT to gain this reputation and as a consequence will build on its existing competitive edge.

4. EMC of large systems

4.1 Background

The European Community Directive for Electromagnetic Compatibility (EMC) [7] that came into force on 1 January 1996 places EMC requirements on large systems such as DFTS. This directive requires all integrators of large

systems to ensure that their system has appropriate EMC. In this instance EMC entails ensuring that the system will not emit radio frequency (RF) signals which could interfere with local systems and will not be affected by the local RF environment (the biggest threat being interference to or from licensed radio services).

There is a fairly straightforward solution for small equipment using standardised test methods written into European normalised standards to allow simple compliance testing and appropriate marking to show conformity (the so-called CE marking). The problem becomes more significant for small systems that will not always fit into a standard EMC test facility such as that operated by TF. It becomes almost impossible for large systems that can only reasonably be integrated together at the operational site. In this case if final testing at the operational site shows a large system to be non-compliant with the EC Directive on EMC, the cost of correcting the problem would generally include a complete strip down and rebuild of the system to implement appropriate EMC fixes.

For complex situations compliance with the EMC Directive can be achieved by the preparation of a technical construction file (TCF). The TCF is a document in which the manufacturer describes their system and why it is compliant with the aims of the EC Directive on EMC. Under this route the system will not necessarily have to meet the standards given for smaller equipment. The technical file route can be used in all cases, but in order to ensure the validity of the statements made, all TCFs must be signed off by a government-designated 'competent body'. Figure 7 describes the routes available to market a product under the terms of the EC Directive on EMC.

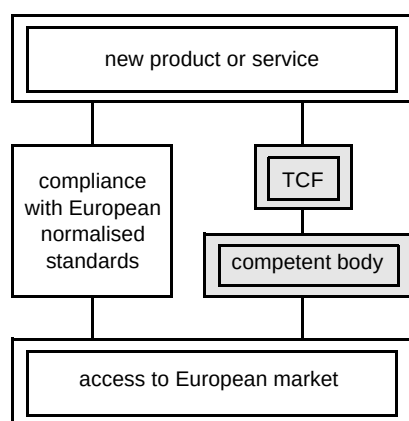


Fig 7 Routes to European market.

A further complication arises for systems upgraded during their life cycle. When any system is enhanced to improve its performance or when new capabilities are added, the upgraded system forms a new system that must meet the requirements of the EC Directive on EMC.

4.2 The risks of non-compliance

Enforcement of the EMC Directive will be complaints driven with penalties including fines, jail sentences and orders banning the use of the equipment for up to six months while tests are completed.

The impact upon BT of non-compliance with the EC Directive on EMC could be significant. The threat of a bespoke customer's network (or indeed of BT's billing system, etc) being disconnected for six months is unacceptable to the business.

The impact of non-compliance needs to be balanced against the costs of *in situ* EMC testing which can be significant.

To address the risk of non-compliance with the Directive, there was a need to develop a laboratory-based method for showing the compliance of a system with the EC Directive on EMC. Ideally this approach would give a method for dealing with upgrades and extensions as well as bespoke and standard systems. It also needed to deal with the frequent situation where simple addition of worst-case emission levels would predict the non-compliance of the system with the EC Directive on EMC.

The solution was the development of a measurement-based modelling system. This could deal with up to a hundred sources and produce a probabilistic analysis of emission levels that would provide realistic results for large systems and which would be sufficiently rigorous to be acceptable both to BT's customers and to the regulatory authority (the Department of Trade and Industry).

The probabilistic nature of the solution allows a limit to be set on the acceptable risk to the business that can be system dependent within the limits acceptable to the selected European competent body.

This modelling would then be the basis for the preparation of TCFs for major systems both for use by BT and provided to external customers.

4.3 EMC mathematical modelling

The difficulty with EMC modelling techniques is the lack of the base data needed to accurately model the EMC performance of a system.

The unique approach taken by the EMC unit of TF has been to review methods for combining measured emission data. The aim is to collate emission results from all the suppliers of equipment within a system and combine them to estimate the actual emission levels.

4.4 Statistical analysis

The simplest approach, currently used by most European test houses, involves summing the emissions from the various pieces of equipment in a scalar fashion. The results do not look acceptable to any system integrator. In simple terms this kind of summation will result in a conclusion of non-compliance for all systems utilising more than two similar PCs because all measured PCs are currently within 6-7 dB of the limit at some frequency. The summation of these emissions will break the limit rendering the system non-compliant with the EC Directive on EMC.

This conclusion is not borne out by an investigation of the emission statistics that show a very limited number of interference cases where the source of the interference is IT equipment. This discrepancy was solved by an investigation of the summation of emissions to find a method that reasonably compared to real life.

The most critical factor in the summation of emissions is the phase relationship of the two emissions. Where the emissions are in phase the system will emit the highest signal. The overall emitted signal will be at a minimum level when the two signals are out of phase.

The most significant information relates to the phase difference between the emissions from two PCs. It rapidly became apparent that the phase is a random variable. There are two principal reasons for this. The equipment is not phase-locked and hence the phase at turn-on is unknown. Even if this phase is locked, the second component of phase, due to the spatial separation between the two emitters, can be known only when the set-up is defined. Any minor movement of the equipment will alter the phase relationship. It is therefore valid to assume that the phase is a random variable, and use a rectangular distribution (all phase angles equally likely) for the purpose of summing emission levels.

In the first instance an algorithm was developed to assess the probability distribution of emission levels for up to nine sources emitting at the same frequency. This was then coded into the so-called 'simple processor'. Verification was a complex challenge for all subsequent models. Only simple systems can be rigorously analysed for their emission levels and even then the uncertainty levels can be significant.

Over a period of three months the TF EMC group has carried out a thorough verification of the model (often against state-of-the-art models developed by the US military). The final results were exceptional. There were no cases in which the model and the measurements did not agree within the experimental uncertainty associated with the measurements.

This investigation led the team to the first real test when an enhanced simple processor (called the 'earth stations processor') was used to assess the emission levels from the

satellite earth stations at Goonhilly, Madley, London Teleport and Aberdeen. The results were then compared with data taken around the periphery of the earth station. The model showed a number of discrepancies and work began to understand and correct for the differences. Ultimately this was shown to be due to two significant factors, multi-path reflections from buildings on the sites and the screening associated with a chain link fence. Once these factors were understood and included in the model, excellent agreement was once again achieved between modelling and measurement. The earth stations were shown to have a probability of conformance well in excess of the 80% acceptability criterion. The TCFs based on this model were issued to the Radio Communications Agency in 1996 and were accepted without challenge. The validated approach has since been used with major success on systems from a 16-card access rack up to BT's DSN switches.

In the past year the model has been used to show the compliance of a variety of systems including BT's ISDN network and 16-card access systems, as well as BT's satellite earth stations. The satellite earth stations' compliance is particularly significant because the modelling work was used to remove an objection to the planning application for an extension to London Teleport.

4.5 Current enhancements

There remained two significant problems:

- the current model is very limited in its capability with respect to cabling,
- it requires significant skill and EMC knowledge for systems with more than nine emitters.

Monte Carlo simulation

The nine-source problem required tens of minutes to assess even simple systems. Simple mathematics shows that this very rigorous 'check every possibility' approach will fail due to time constraints. In 1996 a major code redesign was undertaken by TF with the aim of assessing systems with hundreds of emitters in acceptable times. This work involved redesigning the code to assess the data. In 1996 work commenced on a method, the Monte Carlo simulation, to reduce the calculation time by representative sampling of the probability distribution.

It soon became apparent that this approach offered significant run-time savings. The simulation technique was enhanced and honed until a code was finally developed which can analyse within five minutes a summation of a hundred (or more) sources with the accuracy of the earlier model. This code is currently being validated and verified and the intention is to issue it in 1998.

Wiring problems

Any EMC engineer will tell you that the wiring is the normal source of the emissions from any system. The wiring problem is a particularly complex one because the source is a distributed one and local factors along the cable (e.g. common impedance and differential mode impedance) can significantly change the effect of the cable.

Work has continued throughout this year on a model for the emissions from cabled systems taking into account many of the complexities of the problem (different cable routings, different earthing conditions, different cable types, different terminations, etc). This work is particularly critical for ensuring the compliance of the broadband systems including the very high frequency digital subscriber link (VDSL) system.

This has been a major undertaking and is still continuing with some of the most accurate and meaningful EMC of cabling codes in the world being developed at BT Laboratories. Once complete, the probabilistic approach described above will be added to this model to allow inclusion of cable emissions in the systems processor.

Summary of EMC modelling benefits

While the entirety of the modelling work is not complete, the work to date makes BT a world leader in being able to assess the compliance probability of large systems against international EMC standards. The research carried out by TF is aimed primarily at the emissions from a system, but reciprocity between emissions and immunity allows the code to be used to assess interference into a system. Comparison of these levels with those against which the system has been designed and tested enables the risk of interference with BT systems to be assessed.

These models can also be used to assess changes in EMC performance when systems are changed, upgraded or enhanced and allows the analysis of the risk inherent in any given change.

This has been achieved at a reasonable cost and without the risks associated with *in situ* measurements.

5. Conclusions

The three examples covered in this paper demonstrate how TF has a major role to play in helping BT to succeed in an increasingly hostile environment. The services described here along with those offered by the other units in TF (such as reliability engineering and Internet services) together offer an integrated package of consultancy services to BT businesses.

TF's expertise in ILS was instrumental in helping BT to win the DFTS bid. The ILS skills developed, as well as offering added value to future bids, also offer BT an opportunity to reduce cost by applying the ILS approach to its own internal processes.

TF has also contributed to the management of the business risk associated with IT systems by the development of the end-to-end security process. This will help to reduce overall security costs by building in security from the outset of a new product and system. It will also help to reduce the likelihood and impact, in the form of direct and indirect losses, arising from any actual failure.

The EMC modelling work has made BT a world leader in the assessment of large systems against international EMC standards. These models can also be used to assess changes in EMC performance when systems are changed, upgraded or enhanced, and allow BT to analyse the business risk inherent in any given change.

This paper has shown how, in a competitive environment, TF supports BT in managing the risk of bringing new products and services to market more quickly through an integrated package of world-class consultancy services.

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John Hughes was educated at Manchester Grammar School and Imperial College, London University. He joined BT in 1972 to work on exchange planning. His subsequent career, until 1988, was in Quality Assurance and Electronic Component Reliability.

He then led a programme to develop the capability of BT's equipment suppliers to a point at which the responsibility for component quality and reliability assurance could be safely handed over. From 1992 to 1995 he was manager of the Reliability Engineering Team at the BT Laboratories during which time there was an increasing involvement with reliability modelling to support bids to major account customers.

He is currently the Integrated Logistics Support (ILS) Manager for the Defence Fixed Telecommunications Service contract.



Denis Ubhi joined BT in 1967 and worked on technical adjudication and progress of contracts for telecommunications equipment. He was sponsored to do an MSc in Industrial Mathematics and Statistics at the University of Aston, Birmingham in 1969/70. He is a member of IEEE.

He moved to the Quality and Reliability Centre at Birmingham in 1982, where he was involved in the analysis of reliability data on electronic components and was instrumental in the development of the BT Handbook of Reliability Data. He is also the author of GS5,

the BT Generic Standard for reliability. He moved to BT Laboratories in 1993, where he worked on the DFTS project with a team of reliability and integrated logistics support experts. More recently he has been responsible as the team leader for R&M and ILS tasks.



Martin Wright joined BT in 1987 to manage the electromagnetic compatibility (EMC) test facility and has managed the EMC unit since 1991. He has been deeply involved in the legal and standards activities surrounding the EMC Directive and is specified as a UK competent body for the EMC of telecommunications equipment and ITE.

He currently chairs CISPR/G/WG2 which has responsibility for international standards relating to emissions from large systems.



Paul Akass joined BT in 1975 and has a background in both telecommunications engineering and computing.

Since 1988, he has been working as a senior IT Security consultant, providing security consultancy services to BT major projects and operating units.

He has recently been heavily involved in policy work, and is particularly focused on initiatives which enable IT security as 'business as usual' within the development life cycle. In 1996 he was responsible for

developing the current SPD development process, which has enabled project teams to develop their own SPDs, and has followed up this work on the end-to-end security process.



Edward Chwastek joined BT in 1980 after graduating in Natural Sciences from Cambridge University. His initial work was to evaluate the reliability of thick-film hybrid microcircuits in System-X exchanges. He then moved into the research and development of new techniques for the rapid assessment of the reliability of semiconductor devices. Between 1985 and 1990, he published papers on new methods of reliability assessment, failure analysis techniques and electronic overstress testing of integrated circuits.

From 1990 to 1995 he worked as a business consultant on business re-engineering projects such as the implementation of paperless management systems.

Since 1994, he has been exploiting the use of the intranet for business purposes. His current role is as an intranet consultant in Networks and Systems.